

Working master

GEOMETRY + GQG

Working master

Updated: Sep 11, 2026

CC0 • Anonymous

Start here

This is the current reading entry point for Geometry Maximization, witness-relative geometry, GQG, and the Hidden Quotient. It condenses their common definitions and current status; detailed proofs, instruments, and historical editions remain in the source folder. The “Geometry v2 source” tab preserves this document’s previous body.

The useful common question is: which distinctions must a retained record preserve for the claim or prediction being made? Combining these projects into one document does not identify their state spaces or transfer proof status between them.

Current source map

Witness-relative geometry and non-descent locus: [New geometry](#)

Predictive fibers and minimal-phase companion:

[Geometry Upgrade v0.1 — Predictive Fibers and Minimal Phase](#)

GQG v0.12 instrument and residual closure: [Open source](#)

Hidden Quotient v1.7, qualified theorem and worked restoration: [Open source](#)

Earlier Hidden Quotient native synthesis:

[# THE HIDDEN QUOTIENT — CORE, WITNESS, AND STRIKE](#)

Toroidal working master and validation status: [TOROIDAL — Working Master](#)

Cosmic Time working master: [# Cosmic Time Architecture III · v07](#)

1. Retain the distinction the claim needs

Declare a source domain D , retained map π , and witness W . Work on attained labels $Q=\pi(D)$.

The non-descent locus is

$N_W(\pi)=\{q\in Q : \text{some } x,y\in\pi^{-1}(q) \text{ satisfy } W(x)\neq W(y)\}.$

A same-label pair with different witnesses shows that W cannot be recovered from that label.

The canonical exact witness-resolving refinement is the image of (π,W) , with its projection back to Q . It is coarsest for retaining this witness; it need not be a causal state or a practical predictor.

The no-shape result is essential: without additional assumptions, any subset of Q can occur as the non-descent locus. A local collision does not determine its global geometry. Metric witness diameters support tolerance-dependent questions; a tested pair gives a lower bound.

Openness, regularity, vertical-derivative criteria, and transport require their stated topological, smooth, connected-fiber, and lifting assumptions. Disconnected fibers may conceal discrete

differences. Nonzero witness holonomy implies non-descent under the declared transport; non-descent alone does not establish holonomy.

The ideal string example makes the distinction concrete: at fixed length, (T, μ) and $(cT, c\mu)$ have equal pitch but different tension. Keeping μ repairs tension recovery in that model. It supplies no general mechanism claim.

Arithmetic witness examples

On $y^2 = x^3 + 2x + 2$ over F_{13} , the pairs (P, Q) and $(P, -Q)$, with $P = (2, 1)$ and $Q = (3, 3)$, have retained coordinates $(2, 3)$ but $x(P+Q) = 12$ and $x(P-Q) = 11$. Retaining both y -coordinates recovers the signed inputs and their sum on the declared affine domain; no minimality is claimed. An existential signed-sum relation is a different target from the x -coordinate of that specified sum. In $O = \mathbb{Z}[\sqrt{2}]$, 1 and $u = 1 + \sqrt{2}$ generate the same ideal. Under $\sqrt{2} \mapsto 3$ into F_7 , their base-3 logarithms modulo 3 are 0 and 1. On $D = \{1, u\}$, the $\sqrt{2}$ coefficient c of $L(\alpha) = (\alpha^8 - 1)/3 \bmod 3O$ recovers exactly this target: $W = c$. Recovery on this two-element domain does not establish global recovery on a larger domain.

Both named constants, Belphegor's Prime and $42 \times 10^{42} + 43$, are established primes. The primality qualification concerns what a probable-prime test proves, not the mathematical status of these two integers.

Full domains, arithmetic checks, and corrections to the packet's primality, Semaev,

Weil-descent, and unit-character claims are in New geometry §15: [New geometry](#)

Integration provenance: [# Geometry research packet: comparison and integration review](#) (corrected examples 1–2). The substantive examples remain in New geometry §§15.1–15.2 under the existing framework and claim limits.

Typed calculus and the scope of a repair

New geometry §16 integrates the supplied Typed Defects v0.2 with corrected statements. D1: $N_-(W_1, W_2)(\pi) = N_-W_1(\pi) \cup N_-W_2(\pi)$. D2: $N_-(f \circ W)(\pi) \subseteq N_-W(\pi)$. D3: if $\pi = r \circ \pi'$, then $r(N_-W(\pi')) \subseteq N_-W(\pi)$. D4: $N_-(W|C)(\pi|C) \subseteq N_-W(\pi) \cap \pi(C)$. D5: deterministic autonomy is $N_-(\pi \circ U)(\pi) = \emptyset$.

For a current output B , next-output recovery uses $B \circ U$. These are set-level claims; topological or measurable descent needs the corresponding structure. The five diagnostic patterns are not an exhaustive partition: relation/output loss, unit loss, restriction-relative injectivity, componentwise versus whole-fiber constancy, and missing representation. The last requires an ambient $j: A \rightarrow Z$ and an explicit object outside $j(A)$; enlarging Z alone does not supply it.

The local v2.1 master and both typed-defect formats are now available. Their formulas and organizing labels were reviewed; file availability is not a new empirical result. Detailed corrections and the toroidal application are integrated in their existing documents.

Corrected calculus, §16: [New geometry](#)

Thermodynamic coordination: [# Thermodynamic Coordination + Reflex Geometry · v3](#)

Service and ablation: [Geometry_Direct_Service_Update](#)

Spectral witnesses and foreign imports (v2.2)

The v2.2 increment adds spectral family VI without changing the non-descent test: a computed eta or spectral-flow value alone does not prove information loss. Declare (D, π, W) , the operator or path, and sufficient analytical conventions; then demonstrate variation within a retained-label fiber. Paths may themselves be points of D .

Two exact examples retain identical squared spectra but lose eta, and retain identical endpoint spectra but lose spectral flow. Heat densities can also be witnesses after a domain-to-operator map and evaluation rule are declared. The import table keeps KPZ, Connes–Kreimer, GHY, APS, and the native constructions distinct unless a correspondence is proved. Detailed proofs and sources remain in the existing increment: [GEOMETRY — Upgrade v2.2grok](#)

2. Predictive closure and finite-horizon loss

For deterministic $U:D \rightarrow D$, an autonomous retained update exists exactly when $\pi(x)=\pi(y) \Rightarrow \pi(Ux)=\pi(Uy)$.

For a stochastic kernel P , require equal next-observation laws for every measurable target set whenever the retained labels agree. A transition table fitted to one trace does not establish this universal condition.

For a fixed output B , retained record r , kernel, clock, and horizon h , define $\delta h(z)$ as the supremum of total-variation distances between future B -output laws from states in $r^{-1}(z)$. Increasing h cannot decrease this diameter; refining r cannot increase the diameter within a refined fiber. When $B=r$, zero one-step diameter on every fiber is universal predictive closure.

For a different, coarser B , matching its next output alone does not establish autonomy of r . In deterministic systems, equality of the entire future output sequence defines the coarsest autonomous refinement retaining the output. This characterizes required information; it does not permit future outcomes as predictor inputs.

Worked counterexample: one-step primality is not closure

Let $D=\{n \in \mathbb{Z}: n \geq 2\}$ and $U(n)=2n+1$. Let $q(n)$ indicate primality and let $r(n)=(q(n), q(n) \wedge q(U(n)))$; the second component indicates Sophie Germain primality.

Both 3 and 5 have $r=(\text{true}, \text{true})$. Their successors are 7 and 11, with $r(7)=(\text{true}, \text{false})$ because 15 is composite, and $r(11)=(\text{true}, \text{true})$ because 23 is prime. Thus equal retained states can have unequal next retained states: r is not autonomous. This corrects the research packet's claim that the Sophie Germain label is the required exact predictive refinement.

For a chosen observation q , agreement of $q(U^k(n))$ for every $k \geq 0$ defines a forward-invariant equivalence. It characterizes a predictive quotient without guaranteeing a finite or efficiently computable representation. The exact counterexample is an illustration of this section's criterion, not evidence for a new general closure theorem.

3. Exact 39-screen reference model

Phases are cycles. Set $\alpha=(3-\sqrt{5})/2$ and $\tau=15-39\alpha\approx 0.103325561245899$. Then

$\theta_n=\{\theta_0+n\alpha\}$, $b_n=\lfloor 39\theta_n \rfloor$, $p_n=\{39\theta_n\}$, $s_n=b_n \bmod 3$.

For the departing transition, $\sigma_n=1_{[0,\tau)}(p_n)$:

$p_{n+1}=p_n-\tau+\sigma_n$; $s_{n+1}=s_n-\sigma_n \bmod 3$.

The pair phase $z_n=\{13\theta_n\}=(s_n+p_n)/3$ rotates by $-\tau/3$. Full bins require θ , strand-plus-slip requires z , and slips require p , under the full-circle domain and boundary convention.

The exact count identity is $SN=N\tau+pN-p_0$, hence $|SN-N\tau|<1$. At $\theta_0=0$, the counts are 5/39, 53/507, and 1,034/10,000 departures. Complete slip gaps are 9 or 10. The stationary slip process is nonmixing and has no finite Markov order; the detailed proofs and algebra remain in the source editions.

A smooth conjugacy preserves these observations only when the observations are transported too. Equal-width bins in the new physical coordinate define a different observation. Analytic approximation bounds and finite numerical verification remain distinct from certified floating-point interval computation. The algebraic extensions do not establish a physical central charge.

4. GQG: one operational record, distinct meanings

GQG v0.12 retains separate namespaces for theorem geometry, dyadic admissibility, retained context, and empirical instrumentation. Witness fidelity is claim-relative; fidelity, resolution, and provenance strength are separate properties. A system report records what was said unless its implementation claim is independently supported.

An external residual can establish observable condition dependence under a frozen comparison. It does not reconstruct hidden states or identify the generating mechanism. Source, derived view, detector, decision, action, and outcome remain separately located.

For empirical constraint-response work, determine eligibility before forming the denominator.

Retain Object, Standing, Response, and Route separately. Standing, event compliance, window binding rate, and derived demotion are different quantities. UNKNOWN, PARTIAL, exclusions, and censoring remain explicit. Historical counts cannot become successor rates by relabeling. Confirmatory promotion retains the independent, framework-naive sealed first pass and coordinate-level agreement requirements.

The Open-Seam Atlas records typed partial transitions, provenance, discarded distinctions, correction, alternate routes, and path order. Endpoint equality does not establish path equality or a global manifold.

5. Residual dynamics and the clock correction

The unified record $R=(\delta C, \delta D, \delta K)$ carries chart, dyadic, and retained-context comparisons. Each channel declares its own comparator; subtraction and a common metric are not automatic.

Let qR observe a state pair and Ψ_n advance it. A fixed-time residual update exists only if $qR(z)=qR(z')$ implies $qR(\Psi_n(z))=qR(\Psi_n(z'))$. A single time-homogeneous map additionally

requires agreement across times. Adding a clock can repair cross-time ambiguity; it cannot repair a collision already present at the same time.

UP/DOWN/FLAT/MIXED are coordinate-direction records. Progressive/regressive interpretations require separate polarity and provenance. Residual labels cannot substitute for the empirical eligibility, response, window, and confirmation rules.

6. Hidden Quotient and current limits

The multiplication-representation theorem is relative to the fixed L^2 witness: its kernel consists of locally null functions, and $\|Mf\|$ uses the locally essential supremum. Semifinite reduction and localizable completion are different operations. The v1.7 Radon–Nikodym face retains $v \ll \lambda$ and strong λ -semifiniteness, with $\lambda = \mu_{sf}$. For finite v , λ -semifiniteness still remains; absolute continuity alone does not replace that finite-piece condition. Use the full qualified statement in v1.7 when citing the theorem.

Three worked examples distinguish loss of information from missing representations. A: an infinite-mass point invisible to the multiplication witness becomes visible by adding actual point evaluation, giving $L^\infty([0,1])^{\oplus \mathbb{C}}$ without changing μ . Keep $W(f)=f(p)$ fixed while enlarging π to (π, W) . The alternative singleton encoding loses every point identity, since interval singletons are also null.

B: the full uncountable direct-sum representation is faithful on almost-everywhere classes and localizable despite being non- σ -finite. Restricting observation to countably many slices loses the omitted field. This is a spatial restriction, not a claim about finite temporal memory.

C: counting measure on the countable/co-countable σ -algebra gives an injective but non-surjective inclusion of measurable bounded functions into $\ell^\infty(I) = (\ell^1(I))^*$. A nonmeasurable χ_J defines an omitted functional; finite subset projections converge strongly to its multiplier outside the original algebra. This is a completeness/representability gap. It is not $N_W(\pi)$, which is defined inside the attained image of a total map.

Prediction additionally requires declared dynamics. In the corrected investigation's identity-update example, the partial record is autonomous even though it cannot recover the full field. Swapping an omitted slice into a retained slice breaks that autonomy; retaining both swapped slices repairs it. These are specified illustrative models, separate from the toroidal kernel.

The corrected investigation 1.1 restores the existing v1.7 qualification, supplies the exact examples and counterexamples, and records the repair history. Its edition number does not supersede the v1.7 core. The Seat 2 and Seat 5 readings remain declared interpretations.

Corrected investigation 1.1 — PDF

The separate toroidal kernel has no exact finite-order sector process under its specified unbounded-current, fixed attempted-microtick assumptions. That result does not rule out every finite-dimensional hidden-state representation. A belief state is a formal repair; finite practical compression still needs an output, horizon, tolerance, and error assessment.

The L=3 follow-up reports PASS; the original pilot remains UNRESOLVED and physical Q2 remains NOT_RUN. Chemistry's freeze is unchanged. Global non-descent shape, general holonomy, and physical mechanisms receive no new empirical certification from this consolidation.

7. Service, upkeep, and predictive information

On eligible user-anchored episodes $E=(O,I,A)$, $\pi_DS(E)=(O,A)$. Distinguish A_sel , the selected source text, from A_task , the validated content required by the governing request. Unknown eligibility or completion remains unresolved. Fixed-stage ablation preserves the service class only when the declared object and answer remain equal and the derived presentation is admissible.

The S1–S5, C, and P labels in local v2.1 index existing service results. Correct the minimum's class to $[E]_DS$. Deletion cost is a function of representatives and fails to descend when two same-service representatives have different costs. Historical compliance, provenance, visibility, and necessary dependencies require their own retained witnesses; offline deletion cannot rewrite historical behavior.

Equal served output can conceal different resource margins, capacities, or participation conditions. The product-witness union rule describes that possibility without treating energy, hours, capacity, and consent as one quantity. The thermodynamic companion keeps physical balances, resource bookkeeping, and interpretation separate. No joule, time, or entropy saving follows from deleting a displayed record.

A causal posterior is a sufficient predictive representation, not automatically the coarsest one: different posteriors may predict the same future observations. Finite observed-process Markov order requires factorization of the next-symbol law from observed histories through a fixed finite suffix, under the specified law. A hidden-state collision alone establishes failure of universal lumpability, not every history-level claim. Use the toroidal all-orders proof under its own hypotheses.

The current source statuses remain unchanged: 39-screen algebra, qualified Hidden Quotient v1.7, separate L=3 engineering PASS, original-pilot UNRESOLVED, physical Q2 NOT_RUN, and historical accepted-event discrepancy –1,352. This integration did not replay experiments or supply a global non-descent shape.

Use this master going forward

Edit this first tab for current definitions and status. Keep full proofs, exact schemas, execution receipts, and historical wording in the sources. Record substantive corrections here with the source and scope; create a separate release only when there is a reason to freeze one.

[Project sources and earlier versions](#)

[All active projects](#)

Geometry v2 source

Current integration: finite-horizon loss and toroidal memory

Updated: Sep 8, 2026

This update connects the predictive-closure criterion in §1 to the later toroidal dynamics documents. The v2.0 release description below remains the record of its supplied package; later companion evidence is identified here.

State the prediction target before maximizing retention

Fix a full-state kernel P , a retained record r , an output B , a sampling clock, and a forward-invariant domain D . For $h \geq 1$ define $\delta h(z) = \sup TV(\text{Law}_x(B_1, \dots, B_h), \text{Law}_y(B_1, \dots, B_h))$, with the supremum over $x, y \in D$ satisfying $r(x) = r(y) = z$. Empty or singleton fibers have zero loss. The laws include the declared sampling schedule.

This measures unresolved prediction differences inside one retained fiber. For a fixed output and clock, $\delta h(z)$ is nondecreasing with h because marginalization cannot increase total variation. Refining r shrinks fibers, so the corresponding local diameter cannot exceed its parent-fiber diameter. These statements do not assert a practical optimum or identify a preferred coordinate without an application-specific objective.

When $B=r$, $\delta 1(z)=0$ on every fiber is precisely the universal law-level closure test in §1. When B is a coarser output, zero one-step B loss alone need not make r autonomous. A counterexample pair certifies a lower bound; a finite sample maximum is not a uniform upper bound.

What the toroidal all-orders result settles

For the specified TD-COS-FH-001 kernel at $J=1$, $t=1/2$, $h_6=0$, reference sector $Z000$, cubic $L \geq 2$, and unbounded integer currents, $q(X_{dn})$ has no exact finite Markov order for any fixed integer $d \geq 1$ measured in attempted microticks. Rejections advance that clock. The result covers stationary observation and rules out an exact time-homogeneous finite-order sector process from any initial law under the note's convergence argument.

The proof uses a compressed reversible operator and arbitrarily large admissible current states; it is not an extrapolation from a fitted finite trace. Consequently, no fixed finite suffix of sector labels is an exact autonomous substitute for the declared full state. This does not rule out all

finite-dimensional hidden-state descriptions, estimate practical memory error, or establish mixing speed. Accepted-move clocks, finite-current truncations, and adaptive schedules require their own analysis.

A constructive record that does close

On this countable full-state model, the posterior $v_n(x) = \Pr(X_{dn}=x \mid \text{retained observation history})$, with a specified initial law and Bayesian update, is an autonomous belief-state description. It may require an unbounded record. A practical finite representation needs a declared output, horizon, approximation tolerance, and an independently tested error bound.

Separate validation from scientific conclusions

The later L=3 direct-reference follow-up reports PASS for eight fresh chains and 320,000 retained states, with minimum holonomy count 104 against its threshold 100 and all eight Fourier-sector screens passing. The original pilot remains UNRESOLVED. Raw sample count is not effective sample size. This follow-up does not establish production mixing, a physical confinement result, or Q1/Q2/Q3 collectively; physical Q2 remains NOT_RUN.

Review scope and next useful work

The review checked the sector-count sum and Fourier arithmetic, and compared the character-return formula with exact finite-group convolution in six cases. It did not replay the reported long runs. The all-orders Doc ends mid-code at “assert a”, so its embedded verifier is incomplete even though the written proof is present. Recover the full executable and resolve the accepted-event count mismatch before claiming independent end-to-end reproduction. Then evaluate candidate retained coordinates against a declared finite-horizon prediction target.

Linked sources

Declared toroidal kernel and attempted-microtick schedule: [# Toroidal Dynamics v0](#)

All-orders sector-memory proof:

[# TD-COS-FH-001- the sector process has no finite Markov order](#)

L=3 direct-reference follow-up: [# Tiny-system validation — L=3 direct-reference follow-up v0](#)

Deterministic mismatch investigation:

[# TD-COS-FH-001 — deterministic mismatch investigation v0](#)

Document body

Geometry Maximization v2.0
CC0- NO RIGHTS RESERVED

****Exact predictive phases, a working reference model, and a scoped field-theory update****
6 September 2026

The project now has a compact operational core. The 39-screen has an exact state update, exact finite observations, a proved count bound, and a runnable implementation. Smooth coordinate changes preserve this model when the observations are transported with them. The broader algebra supplies structures on that phase space; its detailed constructions and corrections remain in the preserved v1.6 appendix.

This release consolidates the existing mathematics and adds a reference implementation. It does not report a new physical experiment. Its field-theory changes repair definitions and future reporting requirements; the outstanding empirical questions retain their existing status.

1. The question being solved

An observation is a usable autonomous state only when equal observed states have equal observed successors. For a declared forward-invariant domain $\backslash(D\backslash)$, update $\backslash(U:D\backslash\text{to } D\backslash)$, and observation $\backslash(B\backslash)$, the criterion is

$$\begin{aligned} & \backslash[\\ & B(x)=B(y)\quad\backslash\text{Longrightrightarrow}\quad B(Ux)=B(Uy) \\ & \quad\backslash\text{quad}(x,y\backslash\text{in } D). \\ & \backslash] \end{aligned}$$

It is necessary because a function has one value at each input. It is sufficient because $\backslash(\bar{U}(B(x)):=B(Ux)\backslash)$ is then independent of the chosen representative. The resulting map is unique on $\backslash(B(D)\backslash)$.

When this fails, fix the outputs that must be retained and identify states only when their entire future output sequences agree:

$$\begin{aligned} & \backslash[\\ & x\backslash\text{sim } y\backslash\text{liff } B(U^k x)=B(U^k y)\quad\backslash\text{quad}\backslash\text{text\{for every } } k\backslash\text{ge}0. \\ & \backslash] \end{aligned}$$

This is the coarsest autonomous refinement retaining those outputs. It characterizes the required information; a practical predictor must realize it with currently available coordinates. Future outcomes are not predictor inputs.

For a stochastic full-state kernel (P) , the corresponding universal criterion compares **next-observation laws**:

$$\begin{aligned} &P(x, B^{-1}A) = P(y, B^{-1}A) \\ &\quad \text{whenever } B(x) = B(y), \end{aligned}$$

for every measurable target set (A) . This is strong lumpability. A special stationary initialization can sometimes admit a weaker Markov projection; it does not establish this universal criterion. Finite trace agreement alone establishes neither one. [Detailed proofs and qualifications: v1.6 §§2–3, 13, 20](baseline/GEOMETRY_MAXIMIZATION_v1.6.md).

2. The exact 39-screen

All phases are measured in **cycles**. Define

$$\begin{aligned} &\alpha = \frac{3 - \sqrt{5}}{2}, \quad \quad \quad \\ &\tau = 15 - 39\alpha = \frac{39\sqrt{5} - 87}{2} \\ &\quad = 0.103325561245899 \dots \end{aligned}$$

The full phase and its observations are

$$\begin{aligned} &\theta_n = \{\theta_0 + n\alpha\}, \quad \quad \quad \\ &b_n = \lfloor 39\theta_n \rfloor, \quad \quad \quad \\ &\rho_n = \{39\theta_n\}, \quad \quad s_n = b_n \bmod 3. \end{aligned}$$

Bins are half-open. Every row describes the **departing** transition $(n \text{ to } n+1)$:

$$\begin{aligned} &j_n = (b_{n+1} - b_n) \bmod 39 = 15 - \sigma_n, \\ &\quad \quad \sigma_n = \mathbf{1}_{[0, \tau)}(\rho_n). \end{aligned}$$

In particular, $(\rho_n = \tau)$ gives $(\sigma_n = 0)$; $(\rho_n = 0)$ gives $(\sigma_n = 1)$. The exact reduced updates are

$$\begin{aligned} &\rho_{n+1} = \rho_n - \tau + \sigma_n = \{\rho_n - \tau\}, \quad \quad \quad \\ &s_{n+1} = s_n - \sigma_n \bmod 3. \end{aligned}$$

The pair phase

$$z_n = \{13\theta_n\} = \frac{s_n + \rho_n}{3}$$

satisfies $(z_{n+1} = \{z_n - \tau/3\})$. These follow by writing $(39\theta_n = b_n + \rho_n)$ and using $(39\alpha = 15 - \tau)$.

The required predictive coordinate depends on the retained output:

Output to retain	Minimal phase realization	Time update
Full bin sequence	θ	$\theta \mapsto \theta + \alpha$
Strand and slip together	$z = \{13\theta\}$	$z \mapsto z - \tau/3$
Slip sequence	$\rho = \{39\theta\}$	$\rho \mapsto \rho - \tau$

Minimality here is relative to the entire future of the stated output, on the full circle with the stated boundary convention. It is proved in [v1.6 §§4–6](baseline/GEOMETRY_MAXIMIZATION_v1.6.md), not inferred from the finite runs in this release. Adding the continuous phase repairs predictive closure. Finite labels or a fixed finite slip history do not provide an exact autonomous replacement.

3. Quantities available without statistical fitting

Summing the ρ update gives, for N departures,

$$\begin{aligned} S_N &:= \sum_{n=0}^{N-1} \sigma_n \\ &= N\tau + \rho_N - \rho_0, \\ &\quad |S_N - N\tau| < 1. \end{aligned}$$

This holds for every starting phase. If $\rho_0 = 0$, then $(S_N = \lceil N\tau \rceil)$ for $(N > 0)$.

Start	Departures	Exact slips
$\theta_0 = 0$	39	5
$\theta_0 = 0$	507	53
$\theta_0 = 0$	10,000	1,034

Complete gaps between slip indices have length 9 or 10. Under a uniform starting phase, the exact two-point slip probability is

$$\begin{aligned} &\Pr(\sigma_0 = 1, \sigma_k = 1) \\ &= \max\{0, \tau - |k\tau|_{\mathbb{R}/\mathbb{Z}}\}. \end{aligned}$$

For example, the probabilities at lags 1, 9 and 10 are respectively (0) , $(10\tau - 1)$, and $(1 - 9\tau)$. These are overlap lengths of a rotating interval. They do not assume independent trials.

The stationary slip process is not mixing and is not Markov of any finite order. Its length- (m) word complexity is $(m+1)$; the pair and bin complexities are $(3(m+1))$ and $(39m)$, respectively, for $(m \geq 1)$. An empirical one-step transition table is a description of that lag, not an autonomous process model. The proofs, continued fraction, substitution and return-map calculations remain in [v1.6 §§5–9](baseline/GEOMETRY_MAXIMIZATION_v1.6.md).

4. Geometry changes through a specified coordinate map

For $(|\epsilon| < 1)$, let

$$\begin{aligned} & h_\epsilon(x) = x + \epsilon \frac{\sin(2\pi x)}{2\pi}, \\ & \text{quad } F_\epsilon = h_\epsilon^{-1} R_\alpha h_\epsilon. \end{aligned}$$

Since $(h'_\epsilon = 1 + \epsilon \cos(2\pi x) > 0)$, this is a smooth change of circle coordinates. The phase is $(\theta = h_\epsilon(x))$, and the transported bin is

$$b(x) = \lfloor 39 h_\epsilon(x) \rfloor.$$

All the phase statements above remain exact. The invariant occupation measure in the (x) coordinate is $(h'_\epsilon(x) dx)$.

Equal-width bins in (x) define a different observation. For $(\epsilon = 0.25)$ and $(x_0 = 0)$, the numerical value $(F_\epsilon(0) \approx 0.349737755708554)$ lies in physical bin 13, while its exact phase lies in transported bin 14. A changed observation must be analyzed on its own terms. [v1.6 §25](baseline/GEOMETRY_MAXIMIZATION_v1.6.md).

5. An approximation with an honest error bound

Write $(a(x) = \sin(2\pi x)/(2\pi))$, $(y = x + \alpha)$, and $(d = a(x) - a(y))$. The second-order approximation is

$$G_\epsilon(x) = y + \epsilon d - \epsilon^2 a'(y) d.$$

For degree-one real lifts, the proved uniform residual is

$$\begin{aligned} & |h_\epsilon(G_\epsilon(x)) - h_\epsilon(x) - \alpha| \\ & \leq E_\epsilon, \\ & \text{quad } E_\epsilon = |\epsilon|^3 [c + \pi c^2 (1 + |\epsilon|)^2], \\ & \text{quad } c = |\sin(\pi \alpha)| / \pi. \end{aligned}$$

It implies a one-step coordinate error at most $(E_\epsilon / (1 - |\epsilon|))$ and an accumulated phase error at most $(n E_\epsilon)$ from identical initial data. The derivation is retained in [v1.6 §32.1](baseline/GEOMETRY_MAXIMIZATION_v1.6.md).

For exact computer comparisons, this release uses the slightly larger **rational bound**

$$\begin{aligned} & \widehat{E}_\epsilon \\ & = \frac{|\epsilon|^3}{3} \bigl[1 + (1 + |\epsilon|)^2 \bigr]. \end{aligned}$$

]

Indeed, $\|c\| \leq 1/\pi < 1/3$ and $\|\pi c^2\| \leq 1/\pi < 1/3$, so $\|E_\epsilon\| \leq \widehat{E}_\epsilon$. Rational ϵ makes the new bound exactly representable.

Let $d_n = \min(\rho_n, 1 - \rho_n)/39$ be the exact distance to a bin boundary. The ideal real-valued G has the correct bin whenever

[

$d_n > n \widehat{E}_\epsilon$.

]

Equality is insufficient. At $(n=0)$, identical initial data gives the same label directly; at $(\epsilon=0)$, the ideal approximation is the exact rotation. These two cases also cover initial boundary points.

The implementation distinguishes this analytic certificate from its floating-point coordinate calculation. It has no interval enclosure for trigonometric evaluation or repeated numerical iteration. A tiny numerical error can change a boundary label, including at $(n=0)$. Exact phase labels remain the reference; numerical mismatches remain visible in the outputs.

For (N) labels $(n=0, \dots, N-1)$, the Haar measure of the conservative possible-error cover is at most

[

$\min\{1, 39 \widehat{E}_\epsilon N(N-1)\}$.

]

This is a bound on a set of starting phases, not an observed error rate or a claim of independent errors.

6. What the later algebra adds

The constructions in v1.4–v1.6 act on a specified smooth phase algebra. They explain how symmetry, differentiation, densities and covering maps fit the model.

| Established construction | Practical scope | Detail retained in v1.6 |

|---|---|---|

| Nested phase algebras for degrees 1, 13 and 39 | Exact information reductions; conditional averaging is not multiplication-preserving | §§30–31 |

| Differential algebra, opposite Novikov products and a Gelfand–Dorfman realization |

Operations on smooth phase functions; the naive product fails on the full noncommutative algebra | §35 |

| Witt action and normalized central cocycle | Explicit phase symmetry and degree-dependent covering maps | §36 |

| Tensor-density transport and weighted norms | Coordinate-covariant fields on the phase circle | §39 |

| Integration of a weight-one density cocycle | A specified map to the scalar Virasoro class, including its coboundary | §40 |

| 39-point uniform phase quadrature | Exact trilinear evaluation for input frequencies ≤ 12 ; aliasing beyond that range | §40.4 |

These constructions do not assign a physical central charge or supply a vacuum representation, vertex algebra, knot invariant, or critical exponent for the screen. The surrounding literature stays conditional on its own hypotheses and on a supplied map to this model. [Technical crosswalk](TECHNICAL_APPENDIX_v2.0.md).

7. What carries into the toroidal field theory

The transfer is methodological and definition-specific. Predictive closure requires a declared full state and update; a sector label remains an observable unless it passes the law-level closure criterion.

At zero source, a conserved integer current has integer cut winding. With $\operatorname{div} I = -6n$, the volume average need not be an integer. A cut flux modulo 6 is still well-defined across homologous cuts, and its parity gives the sector. Full periodic ensemble summation projects onto even parity; that projection cannot be read as an observed confinement mechanism.

Formal generator span, realizable paths, observed transitions and mixing are different claims. A state-dependent work stopping rule also needs its own stationarity argument. Missing coverage gives an unresolved closure claim; an exact same-label pair with unequal successor laws proves failure on the stated domain.

The concrete amendments and old-section mapping are in [Field Theory Update v0.3](FIELD_THEORY_UPDATE_v0.3.md). The [Simulation Protocol v0.5](SIMULATION_PROTOCOL_v0.5.md) implements the phase reference lane and specifies future field-lane requirements. Q1, Q2 and Q3 remain separate empirical questions. No field production kernel or physical verdict is implemented in this release.

8. Use and evidence status

Open [the offline explorer](EXPLORER.html) to inspect the included exact runs. It plays back generated data, shows the numerical comparison and exports the trace. Run [the Python reference](simulator/geometry_reference.py) with a supplied configuration to generate a new trajectory; the [README](README.md) gives the command. Inputs are the number of departures, an exact initial phase, and a rational warp parameter. The code uses only the Python standard library.

The executable has a deliberate range: $\{1 \leq N \leq 100\}$, $\theta_0 \in \mathbb{Q}(\sqrt{5}) \cap [0, 1)$, and rational $|\epsilon| \leq 0.99$. The mathematical smooth-conjugacy statement has the larger domain $|\epsilon| < 1$. Floats are not accepted as exact configuration values; use strings such as "0.01" or "1/7".

The [evidence ledger](EVIDENCE_LEDGER_v2.0.json) distinguishes new derivations, proofs retained in the baseline, constructed exact checks, numerical fixtures, conditional references and unresolved claims. Source instructions remain source content. Source hashes identify bytes; they do not certify mathematical or empirical truth.

The next scientific step is a separately specified testable hypothesis, with a declared observation, sampling address, tolerance and evidence gate. This release supplies the reference against which such an experiment can be checked. The historical experiments and the complete v1.6 audit remain preserved in [the baseline archive](baseline/GEOMETRY_MAXIMIZATION_v1.6_verification.zip).